Integrated Security Systems for the CSA Secure Facility

Course: ENGINEER 3PX3: Engineering Economics

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Group: 7

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Group Contribution List

Table 0.0.1 Team member contribution summary

Team Member	Subsystem	Sections Contributed
Jacob Brodersen	Software	Section 3: Economic Evaluation Summary (Software) Section 4: Risk and Uncertainty Analysis (Software) Section 6: Conclusions and Next Steps (Software)
Tridib Banik	Software	Section 1: Introduction and Project Context Section 2: Subsystem Designs and Assumptions (Software) Section 5: Design Recommendations (Software)
Jasvin Mand	Electrical	Section 6 (Electrical): Conclusions and Next Steps Section 2 (Electrical): Subsystem Designs and Assumptions
Shajijan Narendran	Electrical	Section 3: Economic Evaluation Summary (Electrical) Section 4: Risk and Uncertainty Analysis (Electrical)
Zihan Lin	Electrical	Section 5: Design Recommendations (Electrical)

1 Introduction

The Canadian Space Agency (CSA) has commissioned a new secure federal operations facility in Gatineau, Quebec, on publicly owned greenfield land adjacent to existing government infrastructure. The facility will serve as a national command and coordination hub for CSA operations, encompassing satellite communications, data analysis, mission planning, and public engagement. It is assumed to span 20,000 m^2 of floor area, accommodate 150–200 full-time personnel, and operate continuously under high reliability and security standards. The core problem is the need for a reliable, secure, and integrated monitoring system capable of protecting sensitive CSA assets and personnel under strict federal constraints.

This report presents the engineering and economic analysis across two tightly integrated subsystems. The Facility Security Software subsystem (Software team) provides the digital layer of security operations, including monitoring dashboards, entry-point access control, cybersecurity protocols, and automated breach response. The Surveillance and Security Systems subsystem (Electrical team) provides the complementary physical hardware layer, i.e., cameras, biometric scanners, motion detectors, and AI-based monitoring, whose data feeds directly into the software subsystem.

The purpose of this report is to present a final integrated recommendation supported by cost estimates, present worth and annual worth analysis, sensitivity analysis, and risk assessment over a 15-year service life at a 2.4% inflation rate and 7% nominal discount rate. The selected Software design, protecting the facility from data breaches and cybersecurity attacks, yields CA$3,502,083.54 in present worth. The selected Electrical design, providing balanced-cost hardware coverage across 95% of the facility, yields CA$319,979.19.

2 Subsystem Designs and Assumptions

2.1 Electrical Surveillance and Security Systems

The proposed electrical subsystem design contains a comprehensive recording system which provides continuous documentation of all activity within and around the facility and stores all the video surveillance data in a single central database (i.e. a cloud storage system). The system prioritizes both complete spatial coverage through permanently active cameras and sensing devices as well as capital expenditure costs. Video and sensor data can be pulled up from a database instantly, given time and location as input parameters, enabling rapid review and analysis.

Think of our physical security grid as an always-on surveillance network. To ensure 24/7 coverage, the hardware includes 160 PoE cameras (6W each), 300 motion detectors (0.5W each), and 45 biometric access scanners (8W each). All data feeds into a centralized 2.5 kW AI server for rapid threat detection.

Getting this infrastructure installed requires a hefty initial capital expenditure of $348,500. However, running it autonomously generates solid annual savings: $25,000 from dodging major security incidents and $12,000 from reducing system downtime. On the downside, hardware inevitably breaks. We've modelled a 13% annual chance of hardware failure, which could cost about $52,500 per incident for emergency repairs and parts.

2.2 Facility Security Software

The Facility Security Software subsystem provides the digital backbone of the CSA facility's security operations. Its primary role is to protect the facility from data breaches and unauthorized access by integrating camera monitoring, entry-point access control, sensor management, cybersecurity protocols, and automated alert and breach response into a unified software platform. The system interfaces directly with the Surveillance and Security Systems subsystems, consuming data produced by physical cameras and sensors to enable real-time monitoring and response.

Two design alternatives were evaluated. Design A, developed by Tridib Banik, focused on real-time camera control and intuitive dashboards for front-line security personnel, providing essential monitoring functionality at a lower upfront cost. Design B, developed by Jacob Brodersen, expanded upon this by incorporating enterprise-grade encryption, network redundancy, and enhanced cybersecurity defenses, directly addressing the facility's high-security mandate. Design B was selected based on a substantially higher present worth of CA$3,502,083.54 compared to Design A's CA$925,518.83. Although Design A carries a higher standalone IRR (220% vs. 164%), the IRR of the cash flow difference of 150.3%, well above the 10% MARR, confirms the additional investment in Design B is economically justified.

On the digital side, keeping our security software safe from cyberattacks is one of the project's biggest financial wildcards.

- **Mandatory Updates:** There's a 100% chance every year that we'll need to patch vulnerabilities. If a major security gap is found, completely revamping the system could cost up to $200,000.
- **Risk Mitigation:** To control these costs, we're budgeting about $104,224 a year for a dedicated cybersecurity expert. Their regular maintenance should drop the chance of a major overhaul by 25% and cut the potential financial hit in half.
- **System Certification:** The software must pass intense SOC 2 audits, which can drag on for 3 to 12 months and cost anywhere from $10,000 to over $60,000. We've built schedule buffers into our project plan to ensure these audits don't delay the facility's go-live date.

The key design parameters for this subsystem are summarized in the table below:

Table 2.2.1 Software subsystem's key design parameters and their corresponding units.

Parameter	Value	Units
Number of Cameras	323	#
Storage Required	3,431.6	TB
Required Network Bandwidth	149.537	MB/s
Number of Wireless Access Points	5	#

The number of cameras (323) is derived from the 20,000 m² floor area, assuming one camera per 40 m² and a redundancy factor of 1.5. Storage of 3,431.6 TB is based on 40 GB per camera per day at 1080p, retained for one year and fully duplicated for backup. Network bandwidth of 149.537 MB/s is calculated from the per-camera data rate across all 323 cameras. Five wireless access points on WPA3 protocol satisfy the total bandwidth requirement while providing strong encryption and resilience against wireless attacks. Detailed derivations are provided in Appendix A.

2.3 Interface and Aligned Assumptions

To ensure consistency across both subsystems, assumptions were formally aligned through two interface exercises. Table 2.3.1 summarizes all aligned values and their source.

Table 2.3.1. Aligned Assumptions across subsystems from Week 4 and 7 Interface Exercises.

Assumption	Value	Source
Facility floor area	20,000 m^2	Week 4
Camera coverage per camera	40 m^2	Week 4
Camera storage usage	40 GB/day/camera	Week 4
Availability target	99.999%	Week 4
Power outage expectation	5.26 min/year	Week 4
Storage cost	$30/TB	Week 4
Camera unit cost	$100/camera	Week 4
Security guard patrol rate	$35/hour	Week 4
Nominal discount rate	7%	Week 7
Inflation rate	2.4%	Week 7
Service life	15 years	Week 7
Implementation timing	Year 1	Week 7

The Week 4 planning alignment resolved inconsistencies in physical parameters shared between subsystems, ensuring that camera counts, storage sizing, and availability targets were derived from the same facility-level inputs. The Week 7 economic alignment standardized all financial parameters, including the extension of the Software subsystem's service life from 10 to 15 years to match the Electrical subsystem. Both subsystems agreed to avoid double-counting risk-reduction benefits, given that the physical hardware generates the data that enables software-based monitoring. All aligned assumptions are carried forward consistently into the economic analysis presented in Section 3: Economic Evaluation Summary.

3 Economic Evaluation Summary

Deterministic economic evaluation of the software and electrical subsystems utilized PW, AW, and IRR. It is assumed that all costs and benefits are affected by an inflation rate of 2.4%. Inflation was applied by adjusting the real net cash flows to nominal values. The PW and AW were calculated using a nominal annual discount rate of 7% and Excel’s built in NPV and PMT functions. IRR was calculated using the nominal net cash flows and Excel’s IRR function. All calculations/values can be found in the Financial Analysis appendices.

While this summary uses PW, AW, and IRR, these methods have limitations. These do not capture all aspects of the security system’s value. The methods are predicated on every cost and benefit having a direct economic value. As such, soft factors which could impact the economic success of the system (e.g. need/ease of maintenance) are not accounted. These methods have the benefit of providing a concise and relatively meaningful overview of the economic positioning of various designs.

3.1 Software Subsystem

Design A has a PW of $925,518.83, an AW of $131,773.06, and an IRR of 220%. Design B has a PW of $3,502,083.54, an AW of $498,617.91, and an IRR of 164%. Design A has a much lower PW than Design B, indicating superior economic performance of Design B. Design B has much higher upfront costs but also has higher monetized benefits leading to the higher overall PW. The AW results reflect this where Design B has a much larger AW than Design A.

The IRR for both Design A and Design B is higher than our MARR of 10% indicating that both designs are acceptable from a rate of return perspective. For both designs, the IRR value is high due to key high-value monetized benefits: (i) for Design A, avoided breach cost ($50,000), reduced downtime ($60,000), and compliance and penalty avoidance ($50,000), and (ii) for Design B, the replacement of patrol labour ($511,000 per year) and increased security productivity ($245,280 per year). Although the IRR is high, these values still give a meaningful indicator that both designs present positive outcomes for the project. Note that IRR does not agree with the conclusions suggested by PW and AW. Design A has a higher IRR than Design B but has a lower PW and AW. This conflict exists because the ratio between monetized benefits to costs is higher for Design A, but the actual values are much lower than those of Design B.

Overall, the economic evaluation informs us that Design B is the superior economic option. Although Design A has a higher IRR, the magnitude of the cash flows for the project are much lower than Design B; because of this, the PW and AW of Design A are much lower than that of Design B. Overall, we determined that the additional costs we must incur, as well as the lower IRR of Design B, do not outweigh the higher PW and AW (which indicate higher net benefit).

3.2 Electrical Subsystem

For this subsystem, there were three proposed designs: Design A which prioritized low costs, Design B which prioritized long lifespan, and Design C which was a balance of both factors. While Design A offered lowest initial capital expenditure and Design B exhibited the longest service life, it was determined that Design C was economically viable and most profitable in a 15-year window; as a result, it was selected as the primary design (refer to narends_07_T01_Week6.xlsx for Economic Evaluation tables).

After the economic evaluations were conducted, it was found that for Design C, the Present Worth was found to be ~$320,000, the Annual Worth was ~$45,500 and the IRR was 72%. Of the three designs, Design C was the only one to feature a positive PW and AW. These metrics illustrate the feasibility of the proposed security system; this implies that the Design C is the only design which can turn a profit. The IRR is well above the 10% MARR, indicating that the project is acceptable for rate of return.

The IRR increased from 69% before inflation, rising to 72%. This suggests that the monetized benefits are more sensitive to inflationary growth than the operating costs. While the PW and AW for Designs A and B became increasingly negative, both metrics for Design C improved. This occurred because Design C was the only design with an initially profitable cash flow; as a result, the inflationary growth of its benefits successfully outpaced the rising cost of its inputs.

4 Risk and Uncertainty Analysis

4.1 Major Risks and Uncertainty Drivers

The first risk is variability in the operating cost due to fluctuation in usage and pricing of cloud and energy. 84% of organizations struggle to manage cloud spend [1] and data centre energy use is projected to double from 415 TWh (2024) to 945 TWh by 2030 [2]. For a facility such as ours, we estimate an impact of ~\$150K/year, 30% waste = ~\$45,000 with a 75% probability/year (1.33-year annual MTTH).

A second risk is hardware failure or need for immediate maintenance (schedule risk). Drive annual failure rates range from 1.7% - 8% in large cloud deployments [3]. Component-level server replacement (\$15K–\$40K), emergency labour (\$10K–\$15K), and 1–3 days of downtime yields ~\$52,500/incident. We estimate that this risk has a 13% probability per year (7.69-year annual MTTH).

The third risk is uncertainty in time for certifications of cybersecurity for complex systems. If certification is delayed, commissioning/handover can be delayed (schedule risk) [4, 5]. We estimate major delays have a 2% probability/year (50-year annual MTTH) and an impact of \$40,000 [4].

The fourth risk is uncertainty in the updating of cybersecurity. Major gaps/issues demand prompt and extensive revamping of the software system (schedule risk). For a facility such as ours, this will cost ~$200,000 [6]. We estimate that this risk has a 50% probability/year (2-year annual MTTH).

Note: Risk tables can be found in G07_T01_Week9.xlxs.

Table 4.1.1: Risk Matrix

		Impact		
		Low (<$10,000)	Medium ($10,000-$100,000)	High (>$100,000)
Probability	Low (>15 years)		Uncertainty in Time for Certification of Cybersecurity for Whole System (50 years, ~$40,000)	
	Medium (5-15 years)		Hardware Failure (7.7 years, ~$40,000)	
	High (<5 years)		Operating Cost Variability Due to Cloud and Energy Use (1 year, ~$45,000)	Uncertainty in Updating of Cybersecurity (2 year, ~$200,000)

Table 4.1.2 Uncertainty Drivers

Risk Driver	**Uncertainty Range**	**Appropriate Distribution**	**Discussion**
Uncertainty in Updating of Cybersecurity	+-50%	Normal Distribution	This risk follows a normal distribution as there is an average that updating cybersecurity generally costs. However, there are many variables that could increase or decrease the cost of updating the cybersecurity (e.g., new vulnerabilities discovered, new attacks become popular/known to the public, etc.). This leads to extreme cases both where it could cost much more if a large vulnerability is discovered or could cost very little if no major updates are required for the year. Due to this large variance, the cost/impact could easily be greatly increased or decreased depending on the circumstances of the particular year (hence the uncertainty range of +-50%).
Operating Cost Variability Due to Cloud and Energy Use	+-50%	Lognormal Distribution (found from page 26 of lecture 14)	Cloud pricing is positively skewed; costs are bounded at zero but can spike sharply during high-demand periods or after vendor price increases. A Log-Normal distribution captures this asymmetry well. LSTM-based models reduce cloud cost prediction error (MAPE) from 26.7% to 9.8% vs. traditional ARIMA models (WJAETS, 2025: https://journalwjaets.com/sites/default/files/fulltext_pdf/WJAETS-2025-0218.pdf). Flexera (2024) reports 84% of organizations struggle with cloud spend management: https://www.flexera.com/stateofthecloud
Hardware Failure	+-7.5%	Discrete Distribution	In the case of hardware failures, there is a binary outcome for any piece of hardware; therefore, it follows a discrete distribution. As determined earlier, there is a 13% chance for a failure to occur for any piece of hardware.

			Consequently, each individual unit acts as a Bernoulli trial, where the hardware exists in one of two mutually exclusive states: operational (0) or failed (1). Uncertainty stems from the number of pieces of hardware used across the entire system.
Uncertainty in Time for Certification of Cybersecurity for Whole System	+-1%	Uniform Distribution	The certification timeline uncertainty is narrow and bounded at ±1%, with no single value within that range more probable than another. A Uniform Distribution is applied following the GUM (JCGM 100:2008) standard, which states that when only upper and lower bounds are known, Uniform is the appropriate assumption (https://www.bipm.org/documents/20126/2071204/JCGM_100_2008_E.pdf). This reflects a well-controlled certification process with stable regulatory requirements. Mitigation: maintain continuous compliance documentation and conduct pre-certification readiness assessments.

4.2 Risk Mitigation Strategies

Operating Cost Variability Due to Cloud and Energy Usage

To keep costs consistent, reserved cloud pricing, real-time cost monitoring, and quarterly FinOps reviews should be implemented, reducing unexpected cost incidents by 35% [7] and costing ~$6,000 [1].

Hardware Failure

N+1 redundancy, along with scheduled maintenance to reduce the probability of hardware failures, will double procurement costs and increases the labour costs (~$138,248) while decreasing the risk probability from p to p^2 (likelihood becomes ~1.69%).

Uncertainty in Updating of Cybersecurity

A cybersecurity expert should continuously review current attacks/vulnerabilities, perform regular code maintenance, and write documentation / clean code. This would cost ~$104,224/year [8] and reduce this risk's probability by 25% [9].

Uncertainty in Time for Certification of Cybersecurity for Whole System

To reduce scheduling uncertainty, we will perform certification planning, audit readiness checks, freeze system scope during the audit window, and keep buffer time. Implementing this strategy will cost ~$40,000/year [4] and virtually remove the risk [5].

Which Mitigation Strategies to Implement

Reserved pricing, FinOps reviews, and real-time monitoring should be implemented as it delivers a 35% probability reduction at a low cost of ~$6,000/year.

N+1 redundancy should be applied selectively to critical components only, rather than all hardware. This limits capital expenditure while protecting against consequential failure points.

Note that although hiring a cybersecurity expert targets the largest risk-driver, this strategy should not be implemented as costs outweigh the benefits.

4.3 Sensitivity Analysis and Monte Carlo Simulation

For the software subsystem, sensitivity analysis indicates that number of cameras and storage capacity are the most influential variables (steep slope). Concurrently, the design is quite robust in dealing with the required network bandwidth (flat slope). See spider plot (Figure 4.3.1) and G07_T01_Week10_Software_Subsystem.xlxs.

Cybersecurity update uncertainty and operating cost variability due to cloud/energy use dominate economic performance uncertainty, yielding an expected risk-mitigated annual loss of $75,000/year and $33,750/year, respectively. Monte Carlo results (Figure 4.3.2, Table 4.3.1, G07_T01_Week10_Software_Subsystem.xlxs) support this by capturing all risk drivers simultaneously. The stochastic mean PW is significantly lower than the deterministic base case indicating that our design is not robust to uncertainty. Despite this, the 100% probability of PW exceeding zero confirms the software subsystem remains economically viable under uncertainty.

For the electrical subsystem, the sensitivity analysis shows that energy consumption and expected downtime have the biggest impact on economic performance. These trends can be seen in Figure 4.3.3 and G07_T01_Week10_Electrical_Subsystem.xlxs file.

Hardware failure is the most dominant risk with high impact ($52,500/event) and non-negligible probability (13%). Operating cost variability is also important since it has a high probability (75%). The Monte Carlo results (G07_T01_Week10_Electrical_Subsystem.xlxs) shows a stochastic mean PW which is lower than the deterministic base case, suggesting the latter is slightly optimistic and implying that our design is not very robust to uncertainty. However, the probability of PW staying above zero is still close to 100%, which means the system is still economically feasible. The spread in the Monte Carlo results is wider than in the sensitivity analysis, showing that multiple risks happening together can increase overall uncertainty.

Table 4.3.1: Software Subsystem Monte Carlo Results

	PW	AW
Mean	$2,366,434.05	$ 336,926.97
Stdev	$253,547.40	$36,099.45
Probability Value>0:	100.00%	100.00%
Confidence Interval Upper	$2,388,712.09	$ 340,098.86
Confidence Interval Lower	$2,344,156.02	$ 333,755.08

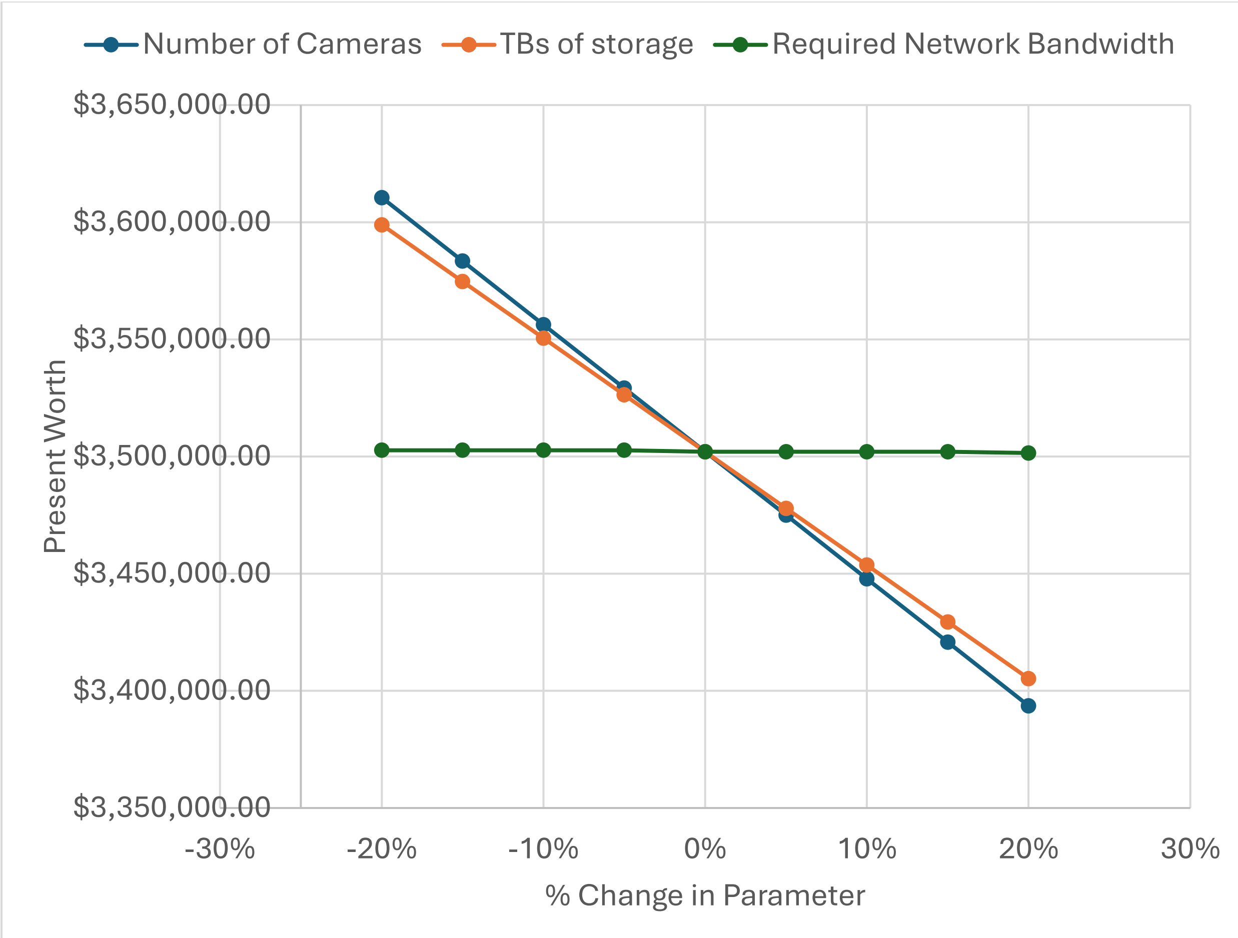

Figure 4.3.1: Software Subsystem Spider Plot

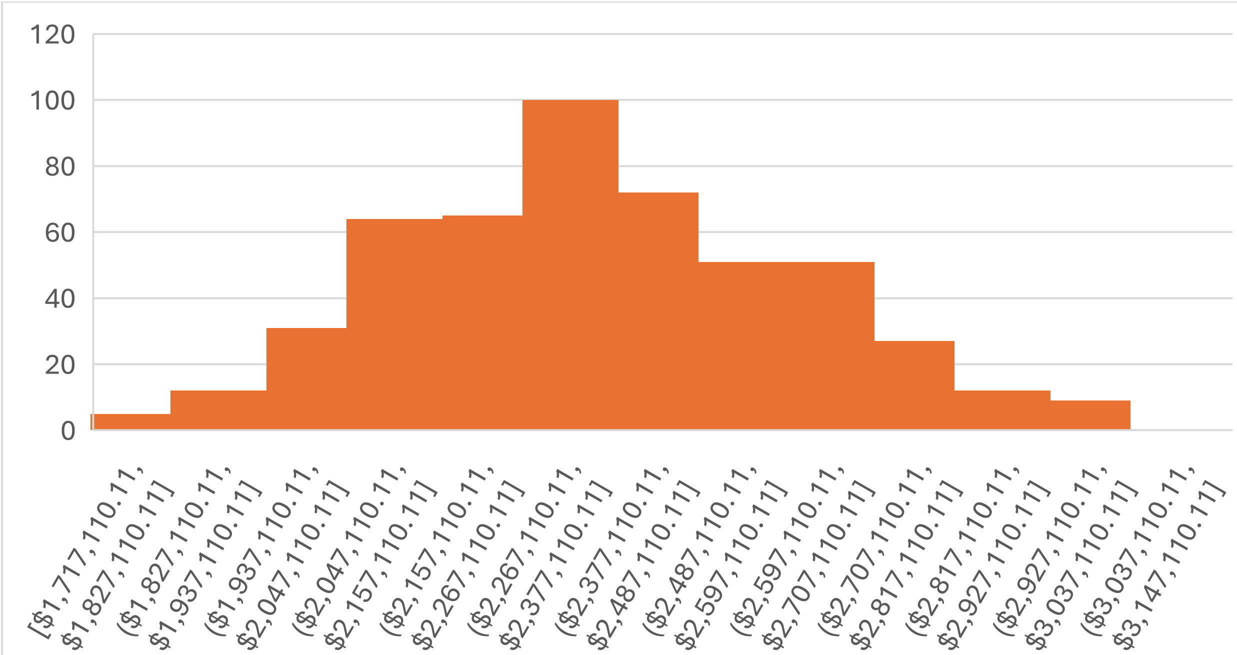

Figure 4.3.2: Software Subsystem Monte Carlo Histogram

Table 4.3.2: Electrical Subsystem Monte Carlo Results

	PW	AW
Mean	$3,062,131.09	$ 435,978.58
Stdev	$21,471.51	$3,057.06
Probability Value>0:	100.00%	100.00%
Confidence Interval Upper	$3,064,017.69	$ 436,247.19
Confidence Interval Lower	$3,060,244.48	$ 435,709.97

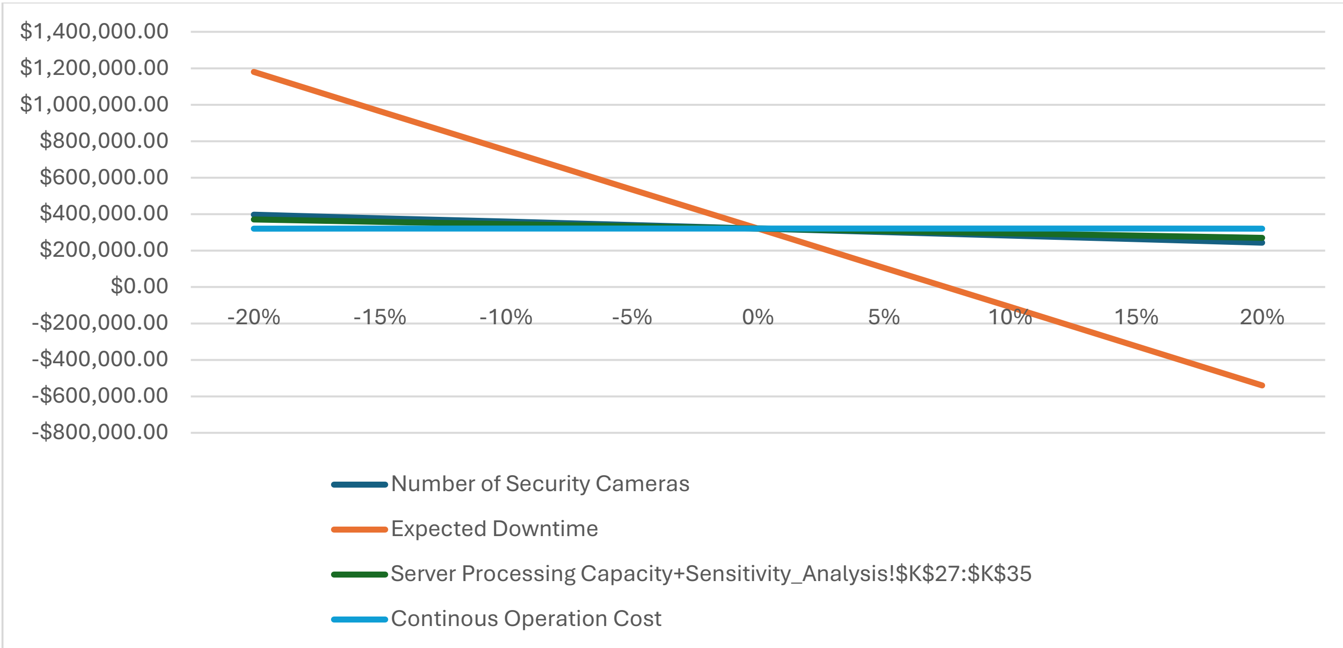

Figure 4.3.3: Electrical Subsystem Spider Plot

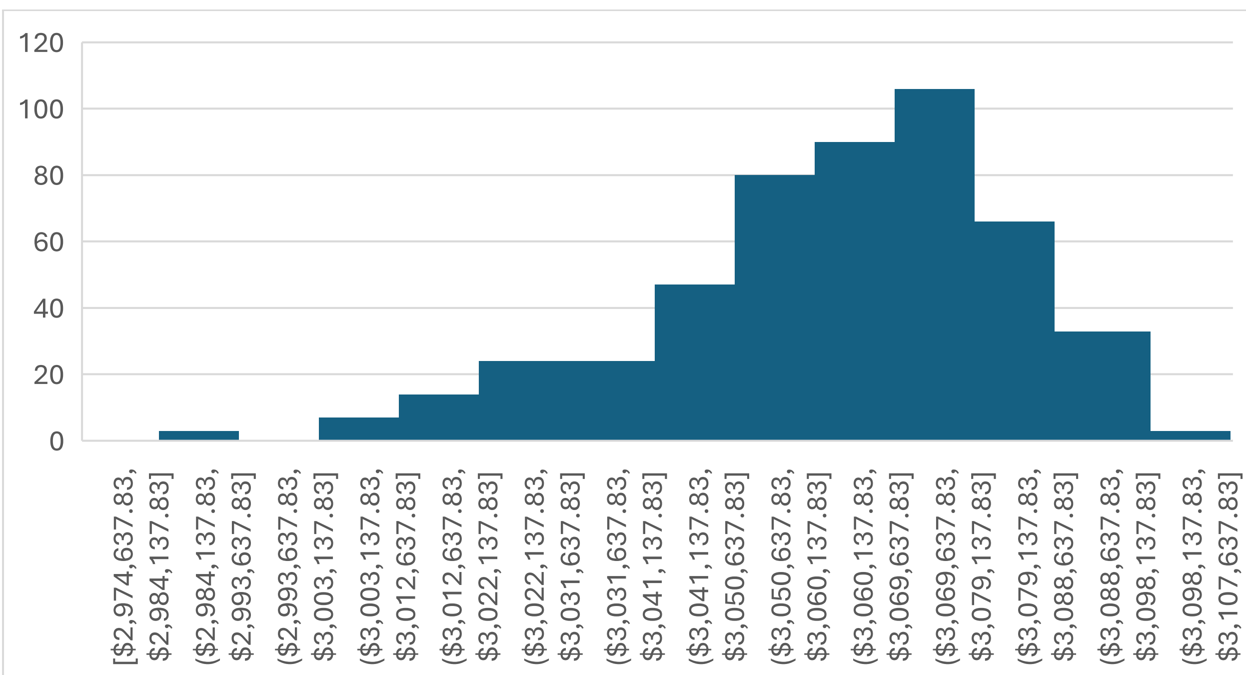

Figure 4.3.4: Electrical Subsystem Monte Carlo Histogram

5 Design Recommendations

5.1 Recommended Design

For the electrical subsystem, Design C is recommended, comprising 160 security cameras, 300 motion detectors, 45 biometric scanners, and a 2.5 kW AI server, with an annual energy consumption of 35,040 kWh/year, a capital cost of approximately $368,500, and an annual operating cost of $24,600. Design C provides facility-wide coverage, eliminates blind spots, strengthens access control, and supports faster threat detection, capabilities essential for a high-security federal facility like the CSA. For the software subsystem, Design B is recommended, implementing a comprehensive monitoring architecture with broader camera coverage, advanced computer vision processing, and enhanced alerting capabilities, making it operationally superior for 24/7 mission-critical operations.

5.2 Justification

Technical feasibility: Design C operates continuously, covers 95% of the facility, and integrates cameras, motion sensors, and biometric scanners across all entry points and interior spaces, fully meeting CSA's operational requirements. Design B satisfies all confirmed software subsystem constraints, fewer than 1,000 sensors, under 100 kW total security power, and 99.9% alert reliability, while delivering stronger monitoring coverage and faster breach response than Design A.

Economic performance: Design C is the only electrical design with a positive Present Worth of $319,979, an Annual Worth of $45,558, and an IRR of 72%, well above the 10% MARR. The alternative electrical designs both produced negative PW values, making them financially unviable. For the software subsystem, under inflation-adjusted conditions (7% nominal discount rate, 2.4% inflation, 10-year horizon), Design B delivers a PW of $3,502,083.54 and AW of $498,617.91, significantly outperforming Design A ($925,518.83 PW, $131,773.06 AW). The incremental IRR of 150.3% far exceeds the 10% MARR, confirming the additional investment is economically justified.

Robustness under uncertainty: Monte Carlo simulation confirms Design C's PW remained positive across all runs, demonstrating resilience against hardware failure and energy cost variability, with N+1 redundancy and regular maintenance as mitigation strategies. For the software subsystem, although Design A carries a higher standalone IRR (220% vs. 164%), Design B's incremental IRR of 150.3% provides a substantial safety margin above the MARR, meaning benefits would need to deteriorate dramatically before Design B becomes unviable. Design A remains a lower-risk fallback if budget constraints tighten.

5.3 Remaining Uncertainties and Limitations

Several uncertainties warrant ongoing attention. Energy costs pose the most immediate risk, since both systems operate continuously, electricity rates rising beyond the assumed 2.4% inflation rate would directly increase OpEx, to which Design B is particularly sensitive. The assumed 15-year hardware lifespan for the electrical subsystem may prove optimistic under heavy use, potentially requiring earlier replacement. Maintenance costs across both subsystems could exceed projections if system complexity leads to more frequent repairs than modelled. Monetized benefits rely on assumed breach frequencies and industry benchmarks rather than facility-specific data, introducing estimation uncertainty. Finally, interface alignment between subsystems requires careful benefit attribution to avoid double-counting security risk reductions. These uncertainties do not undermine either recommendation but highlight the importance of monitoring performance once the integrated system is operational.

6 Conclusions and Next Steps

6.1 Summary

For the software subsystem, Design B offers promising economic performance and satisfies all the identified objectives and constraints. Design B offers robustness to uncertainty, allowing for consistent, positive cash flows with a healthy safety margin. For the electrical subsystem, Design C is the only profitable option. By mixing high-end cameras with cost-effective motion sensors, 95% coverage was achieved while securing a 72% IRR and PW of ~$320,000.

6.2 Next Steps

Future work includes developing and analyzing additional risks of the software subsystem. The current analysis considers only four risks, which is small for a subsystem of this size/importance. A more fulsome consideration of risks is critical to identifying further mitigation strategies. Additionally, a more highly developed plan for the design is required to allow for refinements.

It is also important to identify mission-critical components of the electrical subsystem. This will enable prioritization to improve overall cost-effectiveness in selecting back-up hardware. Once the facility goes live, careful monitoring of real-world energy use and hardware wear-and-tear will be vital to validating assumptions and estimates associated with the project lifespan.

6.3 Trade-offs Reflection

Early-stage design decisions are based upon limited information. Concurrently, it is important to move the project forward and reassure stakeholders of progress. While substantial time can be dedicated to refining estimates to ensure optimal details, there comes a point where the benefits of tweaking assumptions and estimates are outweighed by the costs of not moving forward. This was an important lesson.

Appendices

A.1 Surveillance and Security Systems Design and Financial Analysis (Jasvin Mand)

High-Level Subsystem Design

Think of this design as a comprehensive, fully autonomous security net. The goal here is to provide 24/7 facility-wide surveillance while keeping energy use as efficient as possible. To pull this off, the hardware lineup features 160 PoE cameras drawing just 6W each, 300 motion detectors at a tiny 0.5W, and 45 biometric access scanners using 8W apiece. All of this physical infrastructure feeds directly into a centralized 2.5 kW AI server that processes the data to detect threats instantly and reduce blind spots.

Financial Analysis

Getting this heavy-duty hardware procured, wired, and properly commissioned requires a significant upfront investment—a Year 1 Capital Expenditure (CapEx) of -$368,500. Once the system is live, the ongoing Operating Expenses (OpEx) for electricity, server upkeep, and routine maintenance run about -$24,646 a year.

However, because the system is so efficient and reliable, it generates some serious monetized benefits totaling +$45,324 every single year starting in Year 2. We get these financial wins by avoiding major security incidents ($25,000/yr), reducing expensive facility downtime through faster threat response ($12,000/yr), cutting out false alarms ($6,000/yr), and saving on baseline electricity ($2,278/yr).

Because these annual benefits easily outweigh the yearly operating costs, the system delivers a steady, positive net cash flow of $20,678 each year for the remainder of its 5-year planning cycle.

A.2 Surveillance and Security Systems Design and Financial Analysis (Shajijan Narendran)

High-Level Subsystem Design

For the electrical subsystem, I focused on a surveillance subsystem design which prioritized performance and lifespan over cost. I chose this subsystem because it requires complex trade-offs between high-performance data capture and power infrastructure, involving technical considerations central to electrical load management and high-availability systems. This relates to my discipline of electrical engineering

The problem this subsystem addresses is the need for a ubiquitous documentation system for a 20,000 m² facility. Because the facility handles sensitive information 24/7, traditional event-based recording is insufficient. The main objective of the CSA is to provide a "Total Recall" environment where every square meter is recorded at all times. This ensures that any security incident can be retrospectively analyzed with full context, prioritizing maximum data retention and system lifespan over budget constraints.

Key Design Parameters and Assumptions

Considering my design maximizes performance and reliability, I implemented a strategy which ensured complete coverage of the CSA: 4K-capable hardware is deployed across the entire site, and the system records continuously (24/7) rather than relying on motion triggers. This ensures the system captures the lead-up to any event without sensor lag.

Table A.2.1: Design Parameters

Design Parameter	Value	Unit	Calculations
Central Recording Server Capacity	8.7	PB	Storage Load: 1,200 camera x 240 GB/day/camera (4k) x 30 day retention

Area Under Surveillance	30,000	m^2	Total Site Floor Area, based on data gathered from exisiting space facilities. Assume at least 95% spatial coverage within building is required (excluding washrooms)
Quality & Reliability	99.80%	Success Rate	Biometric Precision: Target success rate for facial/iris recognition in varying luminosity. Requires 4K resolution to achieve the pixel density needed for biometric accuracy.
Surveillance Operation Time	24	hrs/day	Continuous Monitoring: Essential for system availability. 24 hours x 365 days = 8760 hrs/yr
Back-up Security Measures (minutes)	60	minutes	UPS Autonomy: Calculated to allow for "Recovery time after outage" and prevent data loss during power outages. Based on a 45kW system load

These parameters were devised using concepts from Energy Conversion to determine power consumed by electronic devices, Signals & Systems to determine the data rates required for 1080p/4K streams, and Microcontrollers Projects to model the success rate of a controller such as a biometric scanner.

Defined Scope & Boundaries

My subsystem covers the physical surveillance and data storage infrastructure; this includes recording devices such as all fixed and PTZ (Pan-Tilt-Zoom) cameras, storage infrastructure power consumption, and wiring efficiency. The power generation and backup systems (UPS/Generators) are handled by the Electrical Team; my subsystem defines the 40-kW load requirement they must support. The software side (AI analytics, user interface, and alert protocols) is handled by the Software Team. My subsystem provides the high-fidelity data streams and stable storage foundation for their applications.

Financial Analysis

Capital and Operating Cost Assumptions

The financial model for the surveillance subsystem prioritizes long-term reliability and performance over initial cost reduction. The Total CapEx is approximately $1.79 million, primarily driven by high-density hardware deployment and power redundancy for the 20,000 m² facility. Major expenses came from procurement of high-quality devices such as camera hardware, battery systems, and server units. The breakdown of precise cost values can be seen in Table A.2.2. Annual operating costs were set to be constant, and they were periodic in nature. The system generates $76,849 annually in monetized value, largely attributed to system availability and data retention value, alongside minor energy efficiency and emission gains.

Table A.2.2 – CapEx & OpEx

		Year					
	Item	**1**	**2**	**3**	**4**	**5**	
CapEx	Camera Hardware	-$720,000	0	0	0	0	
	Installation Costs	-409,100	0	0	0	0	
	Server Capacity	-302,750	0	0	0	0	
	Backup Battery	-360,000	0	0	0	0	
OpEx	Electricity for server, camera, and scanners	-47,280	-47,280	-47,280	-47,280	-47,280	
	Hardware maintenance Costs	0	-89,592.50	-89,592.50	-89,592.50	-89,592.50	
Benefits	Reduced Emissions	0	48.75	48.75	48.75	48.75	
	Energy Efficiency	0	3000	3000	3000	3000	
	Long Term Data Retention	0	5,000	5,000	5,000	5,000	
	System Availability	0	68,800	68,800	68,800	68,800	**Total**
Net Cash Flow		-1839130	-60023.75	-60023.75	-60023.75	-60023.75	-2079225

Cash Flow Structure

The project utilizes a 15-year lifecycle model. It begins with a significant negative cash flow in Year 1 (-$1,762,281) due to the initial high capital expenditure costs. Years 2 through 15 alternate between positive net cash flows of $29,569 (standard operation) and negative net cash flows of -$60,024 during major maintenance years (Years 4, 8, and 12).

Table A.2.3 – 15-year Cash Flow Table

Year	CapEx	OpEx	Monetized benefits	Net cash flow
1	-$1,791,850	-$47,280	$76,849	-$1,762,281
2	$0	-$47,280	$76,849	$29,569
3	$0	-$47,280	$76,849	$29,569
4	$0	-$136,872.50	$76,849	-$60,024
5	$0	-$47,280	$76,849	$29,569
6	$0	-$47,280	$76,849	$29,569
7	$0	-$47,280	$76,849	$29,569
8	$0	-$136,872.50	$76,849	-$60,024
9	$0	-$47,280	$76,849	$29,569
10	$0	-$47,280	$76,849	$29,569
11	$0	-$47,280	$76,849	$29,569
12	$0	-$136,872.50	$76,849	-$60,024
13	$0	-$47,280	$76,849	$29,569
14	$0	-$47,280	$76,849	$29,569
15	$0	-$47,280	$76,849	$29,569
			PW	-$1,565,587.74
			AW	-$171,893.12

Economic Calculations

The present worth of the project today is **-$1,565,587.74**. The annual worth, or equivalent uniform annual cost of owning and operating the subsystem is **-$171,893.12**. To calculate these values, we used the Net Cash Flow for each year of the project's 15-year lifespan and applied to the PW formula. Then by using a MARR value of 10% and the PW value, the AW was calculated.

Brief Explanation of Results

The financial results indicate that the high-performance surveillance subsystem is not economically viable. Both the Present Worth (PW) and Annual Worth (AW) are negative, which confirms that the total discounted costs of the system exceed its direct monetized benefits over the 15-year lifecycle.

The Internal Rate of Return (IRR) is -72% which is lower than the Minimum Acceptable Rate of Return (MARR), suggesting that the capital allocated to this high-performance design does not generate a positive financial return. This is primarily because a 24/7 security system produces few direct monetized benefits (such as energy savings or emission reductions) compared to the significant CapEx and recurring maintenance costs required to maintain a 99.999% availability target.

A.3 Surveillance and Security Systems Design and Financial Analysis (Zihan Lin)

High-Level Subsystem Design

System Concept and Technology Class

For the electrical subsystem, I focused on surveillance and security systems. I chose this subsystem because it involves many design choices that can be optimized for cost, and it includes technical considerations closely tied to electrical engineering and power optimization.

The problem my subsystem addresses is that the CSA facility needs a dependable surveillance and security system to monitor the building, manage access, and identify potential threats. The facility runs 24/7 and handles sensitive information, so protecting that information is critical.

The main objective of this subsystem is to provide continuous monitoring and access control across the site. The system should be accurate, dependable, and capable of quickly identifying security incidents while minimizing false alarms.

Key Design Parameters and Assumptions

For this design, my priority was to minimize cost while still meeting basic security requirements. I used a risk-based deployment strategy: higher-cost equipment is placed only at high-risk locations, and lower-risk areas rely on motion-triggered monitoring instead of full camera coverage. Video recording is event-driven to reduce storage and infrastructure needs.

Table A.3.1: Key Design Parameters

Design Parameter	Value	Unit	Calculations
Number of security cameras	90	units	Assume total facility floor area ≈ 15,000 m^2 (given project scale). To minimize cost, cameras are not deployed at uniform full-building density; instead, they focus on entrances/exits, corridor junctions, and sensitive areas. Use a cost-minimizing planning density of about 1 camera per 180 m^2. Baseline estimate: 15,000 m^2 ÷ 180 m^2/camera ≈ 83.3 cameras. Round up to cover blind spots and critical points: ≈ 90 cameras.

Average power per camera (PoE)	6	W/camera	Cameras use PoE (Power over Ethernet) and are set to moderate resolution/frame rate (enough for identification at key points without high-end continuous imaging). Typical efficient PoE cameras draw several watts depending on encoding/IR/frame rate; for conservative sizing, choose a mid-range value. Use 6 W per camera as the design-average power for energy and OpEx estimation.
Number of motion detectors	180	units	Motion detectors provide low-cost area monitoring and trigger recording in non-critical spaces. To minimize cost, use a lower-density sensor plan: approximately 1 detector per 100 m² across general areas. Baseline estimate: 15,000 m² ÷ 100 m²/detector = 150 detectors. Add extra detectors for corridors, stairwells, and irregular layout (coverage redundancy): $150 \times 1.2 = 180$ detectors (≈20% contingency).
Average power per motion detector	0.5	W/detector	Assume standard low-power motion sensors (e.g., PIR-based) designed for continuous monitoring. [10] Typical power draw is well below 1 W; choose a

			conservative average value for sizing. Use 0.5 W per detector for total energy and OpEx calculations.
Number of biometric scanners	20	units	To minimize cost, biometric scanners are installed only at high-security access points (not every door). Assume the facility has about 20 high-security entry points (e.g., server rooms, secure labs, restricted storage, control rooms, and staff-only entrances). Therefore, deploy ≈ 20 biometric scanners (≈ 1 per high-security door).

I used concepts from Digital Logic (2DI4), Signals & Systems (3TP3), and Probability & Statistics (3TQ3) to justify this design. Digital logic helped with event-driven decision flows. Signals & Systems informed my choices on data rates and storage needs. Probability & Statistics guided risk-based deployment so that I could achieve acceptable security with fewer devices.

Defined Scope and Boundaries

My subsystem covers the physical security hardware:

Security cameras; Motion detectors and Biometric scanners.

The software side (dashboard, access control logic, alert management) is handled by the Software team. My subsystem provides the data inputs to their system. I am not responsible for software development or cybersecurity protocols.

Financial Analysis

Capital and Operating Cost Assumptions

I split costs into two categories: CapEx (one-time, Year 1) and OpEx (recurring each year). Benefits start in Year 2 after the system is fully operational.

Table A.3.2: Cost Estimation

Cost Driver	Cost Item	Cost Type	Estimation Method	Estimated Value
Number of cameras	Security cameras (90)	CapEx	Three-point estimate: Low \$120, Most likely \$210, High \$300 per camera [11]. Weighted average = (120 + 4×210 + 300)/6 = \$210 per camera. 90 × \$210 = \$18,900	\$18,900
Camera installation	Labour for 90 cameras	CapEx	Professional installation: \$100–\$300 per camera [11]. Assume \$150 per camera for mid-range estimate. 90 × \$150 = \$13,500	\$13,500
Motion detectors	180 sensors	CapEx	Three-point estimate: Low \$50, Most likely \$80, High \$175 per sensor [11]. Weighted average = (50 + 4×80 + 175)/6 = \$86 per sensor. 180 × \$86 = \$15,480	\$15,480

Biometric scanners	20 scanners	CapEx	Market price: $1,500 per unit [12]. 20 × $1,500 = $30,000	$30,000
Electricity	Annual power for cameras and sensors	OpEx	90 cameras × 6W = 540W; 180 sensors × 0.5W = 90W; total 630W = 0.63 kW. Annual consumption: 0.63 kW × 8,760 hrs = 5,518.8 kWh. Ontario electricity rate: $0.13/kWh [13]. 5,518.8 × $0.13 = $717/year	$717/year
Maintenance	Annual hardware upkeep	OpEx	Rough estimate: 5% of equipment cost (cameras + sensors + scanners) = 5% × ($18,900 + $15,480 + $30,000) = $3,219/year	$3,219/year

Total CapEx: $77,880

Total OpEx per year: $3,936/year

Table A.3.3: Benefits Estimation

Benefit Driver	Specific Benefit	How It's Monetized	Estimation	Monetized Benefit

Security guard patrol reduction	Reduced need for physical patrols	Hours saved × wage × number of guards	A security system reduces the need for 2 guards × 8 hours/day × 365 days = 5,840 hours/year. Security guard wage: $20/hour [14] [15]. 5,840 × $20 = $116,800/year	$116,800/year
Incident loss reduction	Fewer theft/security incidents	Baseline annual loss × risk reduction	Baseline crime loss for a facility of this size: estimated $10,000/year based on rising crime trends [16]. Cameras reduce risk by 30%. $10,000 × 30% = $3,000/year	$3,000/year
Downtime avoided	Faster detection → less disruption	Hours avoided × staff × wage	Assume 10 hours/year of downtime avoided × 10 affected staff × $30/hour = $3,000/year	$3,000/year

Compliance risk reduction	Reduced manual audit and reporting effort	Hours saved × wage	Assume 50 hours/year saved × $30/hour = $1,500/year	$1,500/year

Total annual benefits: $116,800 + $3,000 + $3,000 + $1,500 = $124,300/year

Cash Flow Structure

I assumed a 5-year service life for this initial design (to align with the low-cost approach). The system takes one year to implement, so benefits start in Year 2.

Table A.3.4: Annual Cash Flow

Category	Year 1	Year 2	Year 3	Year 4	Year 5
CapEx	-$77,880	$0	$0	$0	$0
OpEx	-$3,936	-$3,936	-$3,936	-$3,936	-$3,936
Benefits	$0	$124,300	$124,300	$124,300	$124,300
Net Cash Flow	-$81,816	$120,364	$120,364	$120,364	$120,364

Brief Explanation of Results

This low-cost design did not turn a profit. The PW and AW are negative, and the IRR is below the MARR of 10%. The main reason is that the benefits ($1,590/year) are too small to recover the upfront cost

($96,570) within the 5-year service life. Even though the design saves on hardware, the limited coverage means fewer monetized benefits.

B.1 Facility Security Software Design and Financial Analysis (Jacob Brodersen)

High-Level Subsystem Design

System Concept and Technology Class

For the Software team subsystem, we chose to focus on the Facility Security Software as outlined in the Week 2 Document. We chose this subsystem as it closely ties in with our Electrical team's choice of Surveillance and Security Systems. Additionally, this subsystem has design decisions that are measurable and can be clearly evaluated. Specifically, the space needed to store the security footage and other related data, network traffic to communicate between the devices, etc. can all be measured both in their own respective units, (e.g., GB for space) as well as in dollar amounts. Also, our subsystem relates to SFWRENG 3RA3 where we learned about cybersecurity. Finally, this subsystem is large enough in scope to produce meaningful costs as described in the following sections.

The problem our subsystem addresses is that there needs to be an efficient method to monitor and enact security within the CSA facility. This includes interfacing with the cameras and security sensors within the facility, as well as other security devices (e.g., building access controls, breach detection, etc.).

Key Design Parameters and Assumptions

This version of the design prioritizes reliability. High reliability was chosen as the design priority because security is a high risk and high importance factor for the CSA (as given in the project description) and as such, the security of the facility should be reliable.

The system includes the following design parameters:

- Number of Cameras: 323
- Storage Required for Camera/Sensor Data: 3,431.6TB
- Uptime Target for Cameras, Sensors, and Security-Related Devices: 99.99999%
- Network Bandwidth for Cameras, Sensors, and Security-Related Devices: 149.537MB/s
- Communication Network Type: WPA3

Key assumptions of this design include:

- Total floor area of facility is 10,000m^2
- A camera is required every 500ft^2
    - Camera coverage must be 1.5 the mathematical required amount for redundancy
- Security footage must be stored for one year
    - Storage must be double the amount mathematically required for redundancy
- Camera is 1080p and runs 24/7, producing 40GB of data per day
- Other security related sensor data as well as devices is magnitudes less than the cameras and therefore can be ignored when measuring value/ impact of the design

Defined Scope and Boundaries of the Subsystem

The scope of the system has two main objectives. The first is to ensure that users have complete discretion over the devices in the subsystem, including both monitoring aspects (cameras and sensors) of the security system, as well as control (e.g., moving cameras, authorizing access to secure rooms, etc.). This includes storing camera and sensor data in case it needs to be accessed later. The second objective is to set up security protocols that, when breached, trigger appropriate alert/breach responses.

For boundaries of the subsystem, we deal with broad costs of setting up the security devices and storage, but we do not account for the intricacies of how to physically set up the security system as this would be part of the electrical subsystem.

Financial Analysis

Capital and Operating Cost Assumptions

The costs and benefits for the design are split up into three categories: CapEx, OpEx, and Monetized Benefits. CapEx are capital expenditures that take place only in the first year. OpEx are operational expenses that occur every year the design is active. Monetized Benefits are benefits that have been translated into dollar amounts and occur every year the design is active, except the first year (as this is the year it takes to get the design active). Below is a table of the identified costs and benefits for this design. Note that full calculations for these values can be found in brodersj_07_T01_Week3.xlxs.

Table B.1.1: Cost and Benefit Table

CapEx	**Value ($)**
Cameras	32,300
Hard Drives	102,948
Wireless Access Points	3,000
OpEx	-
Camera, Sensor, Security-Related Device Maintenance	64,600
Salary of Network Engineer	98,391.67
Data Center Energy	48,728.72
Monetized Benefits	-
Eliminated Patrol Salary	511,000

Expected Value of Avoided Downtime	3,066
Reduced Property Insurance Premiums	1,000
Enhanced Encryption	2,767.53
Security Productivity	245,280

Cash Flow Structure

It is assumed that the design will last ten years, a common lifetime for a security software system [17]. Below are the net cash flow for my design both including inflation and not including inflation.

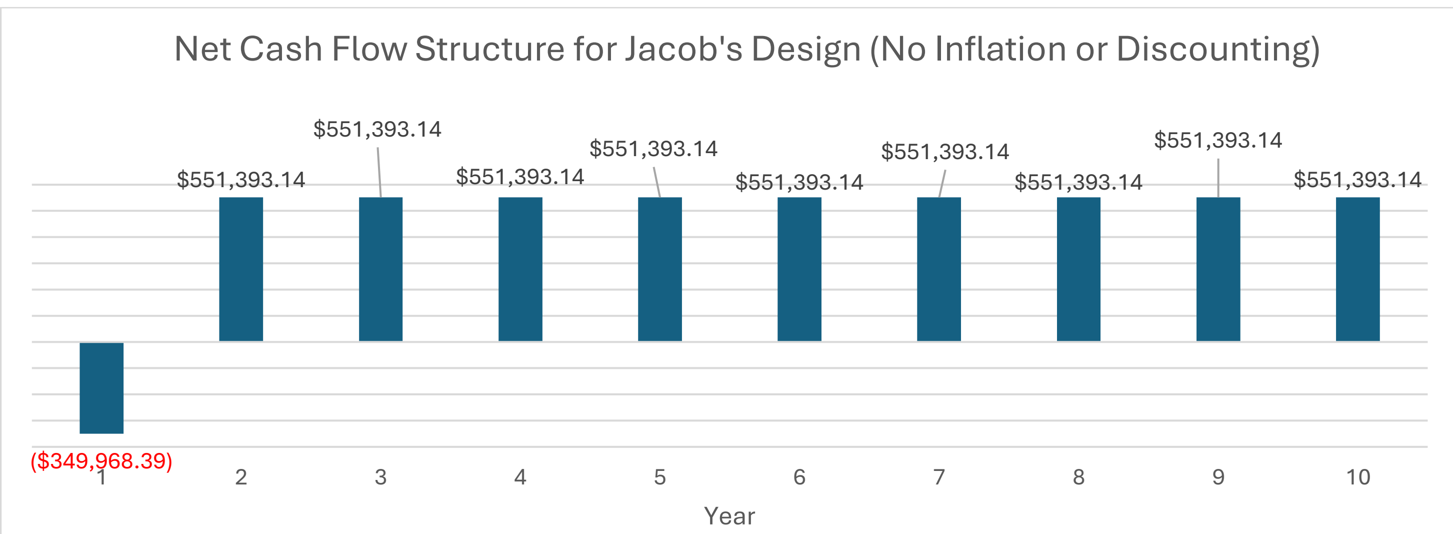

Figure B.1.1: Net Cash Flow for Jacob's Design (No Inflation or Discounting)

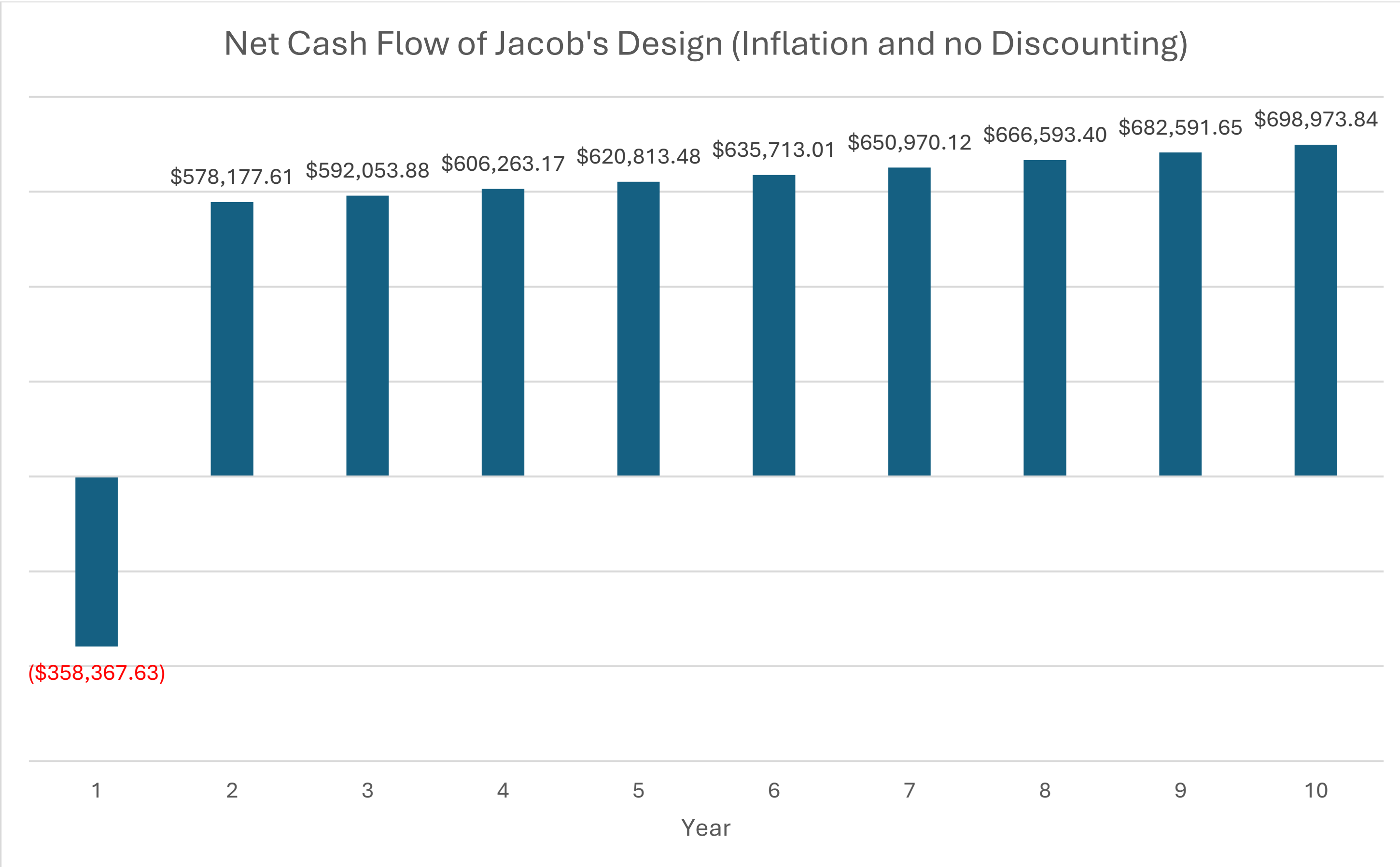

Figure B.1.2: Net Cash Flow of Jacob's Design (Inflation and No Discounting)

Present Worth (PW), Annual Worth (AW), and Other Relevant Economic Calculations Used in the Project

Below are the PW, AW, and IRR calculations for both with inflation and without inflation. Note that IRR is only calculated with inflation. The full, detailed calculations can be found in brodersj_07_T01_Week6.xlxs and brodersj_07_T01_Week7.xlxs.

Table B.1.2: PW and AW Calculation Without Inflation

Year	CapEx	OpEx	Monetized benefits	Net cash flow
1	-$138,248.00	-$211,720.39	$0.00	-$349,968.39

2	$0.00	-$211,720.39	$763,113.53	$551,393.14
3	$0.00	-$211,720.39	$763,113.53	$551,393.14
4	$0.00	-$211,720.39	$763,113.53	$551,393.14
5	$0.00	-$211,720.39	$763,113.53	$551,393.14
6	$0.00	-$211,720.39	$763,113.53	$551,393.14
7	$0.00	-$211,720.39	$763,113.53	$551,393.14
8	$0.00	-$211,720.39	$763,113.53	$551,393.14
9	$0.00	-$211,720.39	$763,113.53	$551,393.14
10	$0.00	-$211,720.39	$763,113.53	$551,393.14
			PW	$3,030,360.73
			AW	$431,455.19

Table B.1.3: PW, AW, and IRR Calculations with Inflation

Year	CapEx (Real)	OpEx (Real)	Monetized benefits (Real)	Net cash flow (Nominal)
1	-$138,248	-$211,720	$0	-$358,367.63
2	$0	-$211,720	$763,114	$578,177.61
3	$0	-$211,720	$763,114	$592,053.88
4	$0	-$211,720	$763,114	$606,263.17
5	$0	-$211,720	$763,114	$620,813.48
6	$0	-$211,720	$763,114	$635,713.01
7	$0	-$211,720	$763,114	$650,970.12

8	$0	-$211,720	$763,114	$666,593.40
9	$0	-$211,720	$763,114	$682,591.65
10	$0	-$211,720	$763,114	$698,973.84
			PW	$3,502,083.54
			AW	$498,617.91
			IRR	164%

Brief Explanation of Results

The PW and AW for the design are positive, indicating that the design would be a net positive (economically). In fact, both PW and AW values are very high, indicating that the design would be significantly economically beneficial. Additionally, the IRR for the project greatly exceeds the given MARR of 10% indicating that it is acceptable in terms of rate of return. Both the CapEx and OpEx are quite high leading to significant loss in Year 1. However, after the first year, these losses are quickly regained due to the significant monetized benefits. As the project progresses, the design becomes more beneficial due to inflation increasing the gap between OpEx and Monetized Benefits.

B.2 Facility Security Software Design and Financial Analysis (Tridib Banik)

High-Level Subsystem Design

System Concept and Technology Class

During Week 2, we decided that, for the Software team subsystem, we would focus on the Facility Security Software. The Facility Security Software is a centralized, event-driven monitoring platform operating 24/7 across the CSA facility. It belongs to the class of real-time supervisory control and data acquisition (SCADA-adjacent) software systems, combining access control enforcement, sensor data validation, computer vision processing, and automated alert management into a single integrated platform.

The system continuously ingests live feeds and sensor signals from the physical infrastructure managed by the Surveillance & Security Systems subsystem. It applies the Bell-LaPadula security model, enforcing "No Read Up, No Write Down", to ensure that camera footage and access logs are only visible to personnel with appropriate clearance levels. A dashboard interface provides authorized security guards with real-time visibility into entry-point statuses, live feeds, and triggered alerts.

Key Design Parameters and Assumptions

The system is designed around the following quantified parameters:

1. **Permanent Data Storage:** 250–300 GB/day, covering continuous recordings from high-security cameras, access logs, metadata, and computer vision event records

2. **CPU Usage:** 120–150 CPU-hours/day at 2.4–3.0 GHz, supporting access control, event aggregation, selective CV processing, and alert generation

3. **Electrical Power Consumption:** 25–30 kW for servers, storage, networking, and redundancy hardware

4. **Video Retention Period:** 60–90 days for critical camera zones, balancing federal compliance with storage cost and data minimization

5. **Cloud Alert Latency:** 1–2 seconds, enabling near-real-time breach detection without compromising operational security

The following table shows how the design parameters were calculated:

Table B.2.1. Quantified Design Parameters for the Facility Security Software Subsystem.

Design Parameter	Value	Unit	Calculations

Permanent data storage	275	GB/day	Median of 250–300 GB/day is 275 GB/day
CPU usage	135	CPU-hours/day	Median of 120–150 CPU-hours/day is 135 CPU-hours/day
Daily Energy Consumption	660	kWh/day	Median of 25-30 kWh = 27.5 kWh. 27.5 kW * 24 hours = 660 kWh/day
Video Retention Period for Critical Cameras	75	days	Median of 60-90 days = 75 days
Cloud latency	1.5	seconds	Median of 1-2 seconds = 1.5 seconds

A key design assumption is that upfront capital costs are minimized by deploying a reduced number of cameras initially, accepting higher operational costs such as additional security personnel to manually cover uncovered zones. All parameters remain within facility-wide constraints: fewer than 1,000 total sensors, fewer than 10 personnel with footage access per week, total security power under 100 kW, and a 99.9% alert system reliability threshold.

Defined Scope and Boundaries of the Subsystem

This subsystem is strictly a software subsystem. Its scope includes:

- Dashboard interfaces for live feed access and entry-point monitoring
- Access control enforcement based on staff and visitor permission levels

- Sensor data validation and camera control logic
- Breach detection and automated alert or alarm triggering

It explicitly excludes all physical hardware, cameras, biometric scanners, motion detectors, and AI monitoring hardware, which fall entirely within the Surveillance & Security Systems subsystem owned by the Electrical Engineering pair. The software subsystem receives data from that hardware layer through defined interfaces but bears no responsibility for hardware installation, calibration, or maintenance. The boundary between the two subsystems is the data interface layer: signals and feeds originating from physical devices enter this subsystem as structured data inputs for processing and visualization.

Financial Analysis

Capital and Operating Cost Assumptions

The financial model separates a one-time capital expenditure from recurring annual operating costs. The sole CapEx item is the initial system integration cost of $25,000 [18]. Annual OpEx totals $41,811, comprising cloud storage ($3,211) [19], compute ($4,900) [20], electricity ($1,200) [21], network/API usage ($2,500) [22], and a SaaS license ($30,000) [23]. Benefits of $184,000/year begin in Year 2 after full adoption, monetized from breach avoidance ($50,000) [24], downtime reduction ($60,000) [25], labour savings ($16,000) [26], compliance penalty avoidance ($50,000) [27], and productivity gains ($8,000) [28]. All real costs and benefits are held constant; inflation is applied nominally at 2.4%/year.

Cash Flow Structure

The analysis spans 10 years at a nominal discount rate of 7% and inflation rate of 2.4%. Real inputs are converted to nominal cash flows using the factor $(1 + 0.024)^t$.

The table below shows annual cash flow from year 1 to 10. Year 2 to 10 have the same CapEx, OpEx, Benefits (monetized) and so net cash flows (total) are all equal, which is why they are not shown.

Table B.2.2. Annual cash flow from year 1 to 10.

Category	Year 1	Year 2	Year 10
Capital Costs (CapEx)	-$25,000	$0	$0
Operating Costs (OpEx)	-$41,811	-$41,811	-$41,811
Benefits (Monetized)	$0	$184,000	$184,000
Net Cash Flow (Total)	-$66,811	$142,189	$142,189

Year 1 is negative due to upfront CapEx and zero benefits during adoption. Years 2–10 produce strong positive flows that grow annually with inflation.

Present Worth (PW), Annual Worth (AW), and Other Relevant Economic Calculations Used in the Project

Without inflation adjustment (nominal discount rare = 7%, 10 years):

PW = $803,349

AW = $114,379/year

Table B.2.3. Present worth and annual worth without considering inflation.

Year	CapEx	OpEx	Monetized benefits	Net cash flow
1	-$25,000.00	-$41,811.00	$0.00	-$66,811.00
2	$0.00	-$41,811.00	$184,000.00	$142,189.00
3	$0.00	-$41,811.00	$184,000.00	$142,189.00
4	$0.00	-$41,811.00	$184,000.00	$142,189.00
5	$0.00	-$41,811.00	$184,000.00	$142,189.00
6	$0.00	-$41,811.00	$184,000.00	$142,189.00

7	$0.00	-$41,811.00	$184,000.00	$142,189.00
8	$0.00	-$41,811.00	$184,000.00	$142,189.00
9	$0.00	-$41,811.00	$184,000.00	$142,189.00
10	$0.00	-$41,811.00	$184,000.00	$142,189.00
			PW	$803,348.93
			AW	$114,378.81

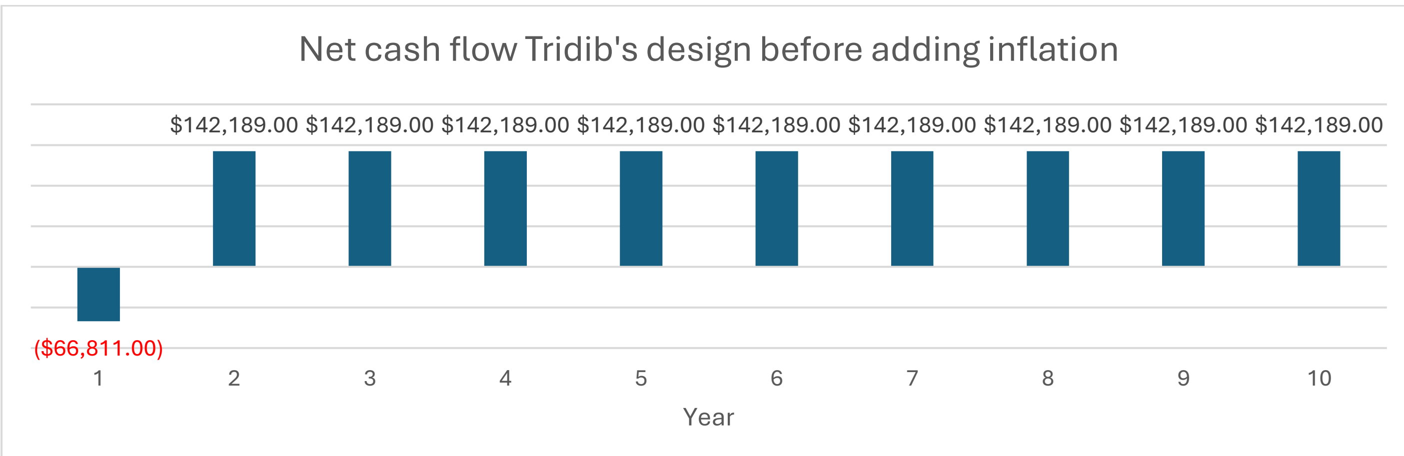

Figure B.2.1. Net cash flow without considering inflation.

With inflation adjustment (nominal discount rate = 7%, inflation rate = 2.4%, 10 years):

PW = $925,519

AW = $131,773/year

IRR = 220%

Table B.2.4. Present worth and annual worth after considering inflation.

Year	CapEx (Real)	OpEx (Real)	Monetized benefits (Real)	Net cash flow (Nominal)

1	-$25,000.00	-$41,811.00	$0.00	-$68,414.46
2	$0.00	-$41,811.00	$184,000.00	$149,095.97
3	$0.00	-$41,811.00	$184,000.00	$152,674.28
4	$0.00	-$41,811.00	$184,000.00	$156,338.46
5	$0.00	-$41,811.00	$184,000.00	$160,090.58
6	$0.00	-$41,811.00	$184,000.00	$163,932.76
7	$0.00	-$41,811.00	$184,000.00	$167,867.14
8	$0.00	-$41,811.00	$184,000.00	$171,895.95
9	$0.00	-$41,811.00	$184,000.00	$176,021.46
10	$0.00	-$41,811.00	$184,000.00	$180,245.97
			PW	$925,518.83
			AW	$131,773.06
			IRR	220%

The IRR of 220% far exceeds the Minimum Acceptable Rate of Return (MARR) of 10%, confirming strong financial viability.

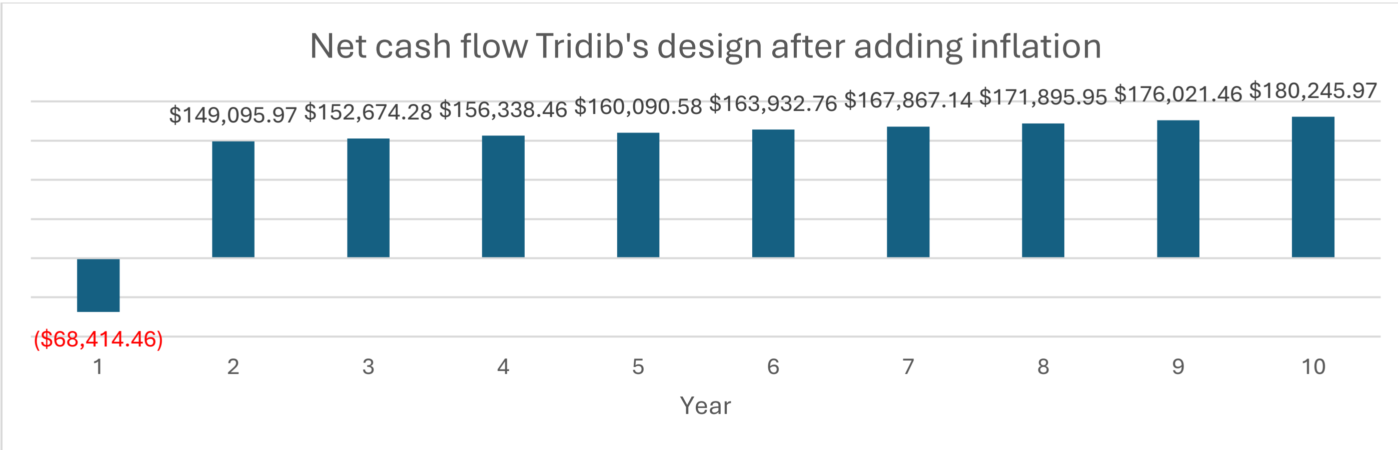

Figure B.2.2. Net cash flow after considering inflation.

Brief Explanation of Results

Both the non-inflation and inflation-adjusted models confirm the subsystem is economically sound. The inflation-adjusted PW of $925,519 is higher than the non-adjusted figure because, while inflation nominally increases both costs and benefits over time, the annual net cash flow is strongly positive (+$142,189/year in real terms), meaning inflated benefits grow by a much larger absolute amount than inflated costs. The dominant cost driver is the SaaS license (72% of OpEx), while the dominant benefit is downtime reduction ($60,000/year). The payback period is approximately 1.46 years, calculated as:

$$\text{Payback Period} = 1 + \frac{|\text{ Cumulative Cash Flow at End of Year 1 }|}{\text{Nominal Cash Flow in Year 2}} = 1 + \frac{68{,}414}{149{,}096} \approx 1.46 \text{ years}$$

Since the cumulative cash flow turns positive partway through Year 2, the system recovers all upfront costs approximately 5.5 months into Year 2. PW and AW are both strongly positive ($925,519 and $131,773/year, respectively), and the IRR of 220% far exceeds the MARR threshold of 10%, strongly supporting investment in the Facility Security Software subsystem.

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